COURSE 2 · ALGEBRA · MODULE 2.3

# Rearranging Formulas

*Printable workbook with worked solutions*

38 problems in seven sets: linear, powers and reciprocals, target twice, exponentials and logs, numbers and units, linearisation and scaling, and spot-the-error. Full worked solutions at the back.

Name _____ Started _____

Before you start

# How to use this workbook

The workbook has 38 problems in seven sets. Sets A to D build the four core techniques; Set E adds numbers and units; Set F applies rearrangement to graphs and scaling; Set G asks you to find mistakes in other people's work. Difficulty is marked with one to three dots.

Work in pencil. Write every step as an operation applied to **both sides**, and check each answer with at least one of the four tests from the Reader (substitution, units, limiting case, scaling) before turning to the solutions, which begin on the page headed *Worked solutions*.

The same problems are available in the interactive workbook, which checks symbolic answers automatically. A good pattern is to attempt a set on paper, then enter your answers online for instant checking.

> **Suggested pacing at four hours a week**
>
> Session 1: listen to the Theory Reader; Sets A and B.
>
> Session 2: Visual Companion and the Untangle bench of the explorer; Sets C and D.
>
> Session 3: Sets E, F and G; the Scale and Linearise benches.
>
> Session 4: redo every problem you got wrong, without looking at the solution, then take the end-of-module quiz.

Set A

# Single occurrence, linear

The target appears once and is only added, subtracted, multiplied or divided. Undo in reverse order.

### **A1** Kinematics: final velocity ●○○

$$v = u + at$$

Make **a** the subject.

### **A2** Straight line ●○○

$$y = mx + c$$

Make **x** the subject.

### **A3** Perimeter of a rectangle ●○○

$$P = 2l + 2w$$

Make **w** the subject.

### **A4** Temperature conversion ●○○

$$F = \tfrac{9}{5}C + 32$$

Make **C** the subject.

### **A5** Distance under uniform acceleration ●●○

$$s = \frac{(u+v)t}{2}$$

Make **u** the subject.

### **A6** Ideal gas law ●○○

$$PV = nRT$$

Make **T** the subject.

Set B

# Powers, roots and reciprocals

The target sits inside a square, a root or a fraction. Isolate the wrapper, then apply its inverse.

### **B1** Kinetic energy ●○○

$$E = \tfrac{1}{2}mv^2$$

Make **v** the subject.

### **B2** Simple pendulum ●●○

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Make **g** the subject.

### **B3** Newtonian gravity ●●○

$$F = \frac{Gm_1m_2}{r^2}$$

Make **r** the subject.

### **B4** Area of a circle ●○○

$$A = \pi r^2$$

Make **r** the subject.

**B5** Thin lens equation ●●●

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Make **v** the subject.

**B6** Volume of a sphere ●●○

$$V = \frac{4}{3}\pi r^3$$

Make **r** the subject.

**B7** Kepler's third law (Newton's form) ●●●

$$T^2 = \frac{4\pi^2}{GM}a^3$$

Make **M** the subject.

Set C

# Target in more than one place

Gather every term containing the target on one side, factor it out, then divide.

### C1 Unknown on both sides ●○○

$$ax + b = cx + d$$

Make **x** the subject.

### C2 Inverting a rational function ●●○

$$y = \frac{x + 2}{x - 3}$$

Make **x** the subject.

### C3 Two resistors in parallel ●●○

$$R = \frac{R_1 R_2}{R_1 + R_2}$$

Make **R2** the subject.

### C4 Doppler effect, moving source ●●●

$$f' = \frac{fv}{v - v_s}$$

Make **vs** the subject.

### C5 Bracketed unknown ●●○

$$a(x - b) = cx$$

Make **x** the subject.

Set D

# Exponentials and logarithms

Isolate the exponential first, then take a logarithm. Logs undo exponents and exponents undo logs.

### D1 Radioactive decay ●○○

$$N = N_0 e^{-kt}$$

Make **t** the subject.

### D2 Acidity ●○○

$$\mathrm{pH} = -\log_{10}[\mathrm{H}^+]$$

Make **H** the subject.

### D3 Compound growth ●●○

$$A = P(1+r)^t$$

Make **t** the subject.

### D4 General exponential ●●○

$$y = ab^x$$

Make **x** the subject.

### D5 Sound level in decibels ●●○

$$L = 10\log_{10}\left(\frac{I}{I_0}\right)$$

Make **I** the subject.

### D6 Newton's law of cooling ●●●

$$T = T_s + (T_0 - T_s)e^{-kt}$$

Make **t** the subject.

Set E

# Numbers and units

Rearrange symbolically first, substitute numbers last, and carry units the whole way.

### **E1** Pendulum length ●○○

A pendulum clock needs a period of exactly 2.00 s where g = 9.81 m/s$^2$. How long must the pendulum be?

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Answer: _____ m

### **E2** Ideal gas temperature ●○○

One mole of gas occupies 0.0224 m$^3$ at 101 325 Pa. Taking R = 8.314 J/(mol·K), what is its temperature?

$$PV = nRT$$

Answer: _____ K

### **E3** Weighing the Earth ●●○

Escape velocity from Earth's surface is 11 186 m/s and Earth's radius is 6.371 × 10$^6$ m. Using G = 6.674 × 10$^{-11}$ N·m$^2$/kg$^2$, find Earth's mass. Give your answer in units of 10$^{24}$ kg.

$$v = \sqrt{\frac{2GM}{R}}$$

Answer: _____ × 10$^{24}$ kg

### **E4** Radiocarbon dating ●●○

Carbon-14 has a half-life of 5730 years, so k = ln 2 ÷ 5730 per year. A bone retains 25% of its original carbon-14. How old is it, in years?

$$N = N_0 e^{-kt}$$

Answer: _____ years

### **E5** Image distance ●○○

An object sits 0.30 m from a converging lens of focal length 0.10 m. Where does the image form?

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Answer: _____ m

### **E6** Doubling time ●○○

A population grows at 2% per year. How many years until it doubles?

$$A = P(1 + r)^t$$

Answer: _____ years

Set F

# Linearisation and scaling

Rearrange a law into the shape y = mx + c to read constants off a graph, or to predict how one quantity scales with another.

### F1 Linearising the pendulum ●●○

You time a pendulum at several lengths L. Which graph gives a straight line through the origin, and what is its gradient?

$$T = 2\pi\sqrt{\frac{L}{g}}$$

**A.** T against L; gradient $2\pi/g$

**B.** $T^2$ against L; gradient $4\pi^2/g$

**C.** T against $\sqrt{L}$; gradient $g/(2\pi)$

**D.** log T against L; gradient ½

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### F2 Linearising decay ●●○

Counts N from a radioactive source are recorded over time t. What should you plot to get a straight line?

$$N = N_0 e^{-kt}$$

**A.** N against t

**B.** ln N against t; gradient −k, intercept $\ln N_0$

**C.** N against ln t

**D.** 1/N against t

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### F3 Scaling an orbit ●●○

Kepler's third law says $T^2$ is proportional to $a^3$. If a planet's orbital radius were four times larger, by what factor would its period grow?

$$T^2 \propto a^3$$

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Answer: _____ ×

### F4 Inverse-square scaling ●○○

Two masses are moved three times further apart. By what factor does the gravitational force change? Give a decimal.

$$F = \frac{Gm_1m_2}{r^2}$$

Answer: _____ ×

### F5 Power laws ●●●

You suspect y = a $x^n$ but do not know n. What plot reveals n?

$$y = ax^n$$

**A.** y against x; gradient n

**B.** log y against x; gradient n

**C.** log y against log x; gradient n

**D.** y against log x; gradient a

Set G

# Spot the error

Each worked attempt contains one mistake. Find it.

### G1 Kinetic energy ●○○

A student solves E = ½mv² for v and writes v = √(2E) ÷ m. What went wrong?

$$E = \tfrac{1}{2}mv^2$$

**A.** Nothing; the answer is correct

**B.** The m should be inside the square root

**C.** The 2 should be outside the root

**D.** They should have halved E, not doubled it

### G2 Thin lens ●●○

From 1/f = 1/u + 1/v, a student 'flips everything' and writes f = u + v. Why is this wrong?

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

**A.** It is right; reciprocals can always be taken term by term

**B.** The reciprocal of a sum is not the sum of the reciprocals

**C.** They forgot a minus sign

**D.** f should be squared

### G3 Square roots ●○○

Solving $x^2 = 9$, a student writes x = 3. What is missing?

$$x^2 = 9$$

**A.** Nothing

**B.** The solution x = −3

**C.** The solution x = 9

**D.** The solution x = 1/3

Answers

# Worked solutions

Every solution shows one clean route. Other correct routes exist; if yours differs, check that your final expression agrees with this one by substituting a few numbers.

## Set A · Single occurrence, linear

### A1 Kinematics: final velocity

| | | |
|---|---|---|
| 1 | Subtract u from both sides to remove the added term. | $v - u = at$ |
| 2 | Divide both sides by t. | $a = \frac{v-u}{t}$ |

ANSWER

$a = \frac{v-u}{t}$

### A2 Straight line

| | | |
|---|---|---|
| 1 | Subtract c from both sides. | $y - c = mx$ |
| 2 | Divide by m. | $x = \frac{y-c}{m}$ |

ANSWER

$x = \frac{y-c}{m}$

### A3 Perimeter of a rectangle

| | | |
|---|---|---|
| 1 | Subtract 2l from both sides. | $P - 2l = 2w$ |
| 2 | Divide by 2. | $w = \frac{P-2l}{2}$ |

ANSWER

$w = \frac{P-2l}{2}$

### A4 Temperature conversion

| | | |
|---|---|---|
| 1 | Subtract 32 (the last thing done to C). | $F - 32 = \frac{9}{5}C$ |
| 2 | Multiply by five-ninths, the reciprocal of nine-fifths. | $C = \frac{5(F-32)}{9}$ |

ANSWER

$C = \frac{5(F-32)}{9}$

### A5 Distance under uniform acceleration

| | | |
|---|---|---|
| 1 | Multiply both sides by 2. | $2s = (u + v)t$ |
| 2 | Divide by t. | $\frac{2s}{t} = u + v$ |
| 3 | Subtract v. | $u = \frac{2s}{t} - v$ |

ANSWER

$u = \frac{2s}{t} - v$

### A6 Ideal gas law

| | | |
|---|---|---|
| 1 | T is multiplied by n and R. Divide both sides by nR. | $T = \frac{PV}{nR}$ |

ANSWER

$T = \frac{PV}{nR}$

## Set B · Powers, roots and reciprocals

### B1 Kinetic energy

| | | |
|---|---|---|
| 1 | Multiply both sides by 2. | $2E = mv^2$ |
| 2 | Divide by m. | $\frac{2E}{m} = v^2$ |
| 3 | Take the square root. Speed is positive here, so keep the positive root. | $v = \sqrt{\frac{2E}{m}}$ |

ANSWER

$v = \sqrt{\frac{2E}{m}}$

### B2 Simple pendulum

| | | |
|---|---|---|
| 1 | Divide by 2 pi to isolate the root. | $\frac{T}{2\pi} = \sqrt{\frac{L}{g}}$ |
| 2 | Square both sides. | $\frac{T^2}{4\pi^2} = \frac{L}{g}$ |
| 3 | Take reciprocals of both sides so g is on top. | $\frac{4\pi^2}{T^2} = \frac{g}{L}$ |
| 4 | Multiply by L. | $g = \frac{4\pi^2 L}{T^2}$ |

ANSWER

$g = \frac{4\pi^2 L}{T^2}$

## **B3** Newtonian gravity

| | | |
|---|---|---|
| 1 | Multiply both sides by r squared. | $Fr^2 = Gm_1m_2$ |
| 2 | Divide by F. | $r^2 = \frac{Gm_1m_2}{F}$ |
| 3 | Take the positive square root (r is a distance). | $r = \sqrt{\frac{Gm_1m_2}{F}}$ |

ANSWER

$$r = \sqrt{\frac{Gm_1m_2}{F}}$$

## **B4** Area of a circle

| | | |
|---|---|---|
| 1 | Divide by pi. | $\frac{A}{\pi} = r^2$ |
| 2 | Take the positive square root. | $r = \sqrt{\frac{A}{\pi}}$ |

ANSWER

$$r = \sqrt{\frac{A}{\pi}}$$

## **B5** Thin lens equation

| | | |
|---|---|---|
| 1 | Subtract one over u to isolate the v term. | $\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$ |
| 2 | Combine the right side over the common denominator uf. | $\frac{1}{v} = \frac{u-f}{uf}$ |
| 3 | Take the reciprocal of both sides. This is allowed only once each side is a single fraction. | $v = \frac{uf}{u-f}$ |

ANSWER

$$v = \frac{uf}{u-f}$$

## **B6** Volume of a sphere

| | | |
|---|---|---|
| 1 | Multiply by 3 and divide by 4 pi. | $\frac{3V}{4\pi} = r^3$ |
| 2 | Take the cube root. Cube roots have only one real value, so no sign choice is needed. | $r = \sqrt[3]{\frac{3V}{4\pi}}$ |

ANSWER

$$r = \sqrt[3]{\frac{3V}{4\pi}}$$

### **B7** Kepler's third law (Newton's form)

| | | |
|---|---|---|
| 1 | Multiply both sides by GM to clear the denominator. | $GMT^2 = 4\pi^2 a^3$ |
| 2 | Divide by G T squared. | $M = \frac{4\pi^2 a^3}{GT^2}$ |

ANSWER

$M = \frac{4\pi^2 a^3}{GT^2}$

## Set C · Target in more than one place

### **C1** Unknown on both sides

| | | |
|---|---|---|
| 1 | Move every x term to the left and every other term to the right. | $ax - cx = d - b$ |
| 2 | Factor out x. | $x(a - c) = d - b$ |
| 3 | Divide by a minus c (valid when a is not equal to c). | $x = \frac{d - b}{a - c}$ |

ANSWER

$x = \frac{d - b}{a - c}$

### **C2** Inverting a rational function

| | | |
|---|---|---|
| 1 | Multiply both sides by x minus 3. | $y(x - 3) = x + 2$ |
| 2 | Expand. | $xy - 3y = x + 2$ |
| 3 | Collect x terms on the left. | $xy - x = 3y + 2$ |
| 4 | Factor out x. | $x(y - 1) = 3y + 2$ |
| 5 | Divide by y minus 1. | $x = \frac{3y + 2}{y - 1}$ |

ANSWER

$x = \frac{3y + 2}{y - 1}$

### **C3** Two resistors in parallel

| | | |
|---|---|---|
| 1 | Multiply by the denominator. | $R(R_1 + R_2) = R_1 R_2$ |
| 2 | Expand. | $RR_1 + RR_2 = R_1 R_2$ |
| 3 | Collect R2 terms on one side. | $RR_1 = R_1 R_2 - RR_2$ |
| 4 | Factor out R2. | $RR_1 = R_2(R_1 - R)$ |
| 5 | Divide. | $R_2 = \frac{RR_1}{R_1 - R}$ |

ANSWER

$R_2 = \frac{RR_1}{R_1 - R}$

### C4 Doppler effect, moving source

| | | |
|---|---|---|
| 1 | Multiply both sides by v minus v s. | $f'(v - v_s) = fv$ |
| 2 | Expand. | $f'v - f'v_s = fv$ |
| 3 | Move the v s term to one side. | $f'v - fv = f'v_s$ |
| 4 | Factor v and divide by f prime. | $v_s = \frac{v(f' - f)}{f'}$ |

ANSWER

$$v_s = \frac{v(f' - f)}{f'}$$

### C5 Bracketed unknown

| | | |
|---|---|---|
| 1 | Expand the bracket. | $ax - ab = cx$ |
| 2 | Collect x terms. | $ax - cx = ab$ |
| 3 | Factor and divide. | $x = \frac{ab}{a - c}$ |

ANSWER

$$x = \frac{ab}{a - c}$$

## Set D · Exponentials and logarithms

### D1 Radioactive decay

| | | |
|---|---|---|
| 1 | Divide by N nought to isolate the exponential. | $\frac{N}{N_0} = e^{-kt}$ |
| 2 | Take the natural log of both sides. | $\ln\left(\frac{N}{N_0}\right) = -kt$ |
| 3 | Divide by minus k. Flipping the fraction inside the log absorbs the minus sign. | $t = \frac{\ln(N_0/N)}{k}$ |

ANSWER

$$t = \frac{\ln(N_0/N)}{k}$$

### D2 Acidity

| | | |
|---|---|---|
| 1 | Multiply by minus one. | $-\mathrm{pH} = \log_{10}[\mathrm{H}^+]$ |
| 2 | Raise 10 to the power of each side. Ten-to-the-power undoes log base ten. | $[\mathrm{H}^+] = 10^{-\mathrm{pH}}$ |

ANSWER

$$[\mathrm{H}^+] = 10^{-\mathrm{pH}}$$

### D3 Compound growth

| | | |
|---|---|---|
| 1 | Divide by P. | $\frac{A}{P} = (1+r)^t$ |
| 2 | Take logs of both sides. The log of a power brings the exponent down. | $\ln\left(\frac{A}{P}\right) = t\ln(1+r)$ |
| 3 | Divide by the log of one plus r. | $t = \frac{\ln(A/P)}{\ln(1+r)}$ |

ANSWER

$$t = \frac{\ln(A/P)}{\ln(1+r)}$$

### D4 General exponential

| | | |
|---|---|---|
| 1 | Divide by a. | $\frac{y}{a} = b^{\,x}$ |
| 2 | Take logs. | $\ln\left(\frac{y}{a}\right) = x\ln b$ |
| 3 | Divide by ln b. | $x = \frac{\ln(y/a)}{\ln b}$ |

ANSWER

$$x = \frac{\ln(y/a)}{\ln b}$$

### D5 Sound level in decibels

| | | |
|---|---|---|
| 1 | Divide by 10. | $\frac{L}{10} = \log_{10}\left(\frac{I}{I_0}\right)$ |
| 2 | Raise 10 to each side. | $10^{L/10} = \frac{I}{I_0}$ |
| 3 | Multiply by I nought. | $I = I_0 \cdot 10^{L/10}$ |

ANSWER

$$I = I_0 \cdot 10^{L/10}$$

### D6 Newton's law of cooling

| | | |
|---|---|---|
| 1 | Subtract T s, the surroundings temperature. | $T - T_s = (T_0 - T_s)e^{-kt}$ |
| 2 | Divide by the bracket to isolate the exponential. | $\frac{T-T_s}{T_0-T_s} = e^{-kt}$ |
| 3 | Take natural logs. | $\ln\left(\frac{T-T_s}{T_0-T_s}\right) = -kt$ |
| 4 | Divide by minus k and flip the fraction to absorb the sign. | $t = \frac{1}{k}\ln\left(\frac{T_0-T_s}{T-T_s}\right)$ |

ANSWER

$$t = \frac{1}{k}\ln\left(\frac{T_0-T_s}{T-T_s}\right)$$

## Set E · Numbers and units

## E1 Pendulum length

1 Rearrange for L first. $L = \frac{gT^2}{4\pi^2}$

2 Substitute. $L = \frac{9.81 \times 2.00^2}{4\pi^2} = \frac{39.24}{39.48}$

3 Evaluate. $L \approx 0.994 \text{ m}$

ANSWER 0.994 m

## E2 Ideal gas temperature

1 Rearrange. $T = \frac{PV}{nR}$

2 Substitute. $T = \frac{101325 \times 0.0224}{1.00 \times 8.314}$

3 Evaluate. This is 0 °C, the familiar standard condition. $T \approx 273 \text{ K}$

ANSWER 273 K

## E3 Weighing the Earth

1 Square both sides. $v^2 = \frac{2GM}{R}$

2 Rearrange for M. $M = \frac{v^2 R}{2G}$

3 Substitute. $M = \frac{(11186)^2 \times 6.371 \times 10^6}{2 \times 6.674 \times 10^{-11}}$

4 Evaluate. $M \approx 5.97 \times 10^{24} \text{ kg}$

ANSWER $5.972 \times 10^{24}$ kg

## E4 Radiocarbon dating

1 Rearrange (see D1). $t = \frac{\ln(N_0/N)}{k}$

2 N nought over N is 1 divided by 0.25, which is 4. $t = \frac{\ln 4}{\ln 2/5730}$

3 Since ln 4 = 2 ln 2, this is exactly two half-lives. $t = 2 \times 5730 = 11\,460 \text{ years}$

ANSWER 1.146e+04 years

## E5 Image distance

1 Use the rearrangement from B5. $v = \frac{uf}{u - f}$

2 Substitute. $v = \frac{0.30 \times 0.10}{0.30 - 0.10} = \frac{0.030}{0.20}$

3 Evaluate. $v = 0.15 \text{ m}$

ANSWER 0.15 m

### E6 Doubling time

1 Doubling means A over P equals 2. Use the rearrangement from D3. $t = \frac{\ln 2}{\ln 1.02}$

2 Evaluate. Compare the rule of 70: 70 divided by 2 gives 35. $t \approx 35.0 \text{ years}$

ANSWER 35 years

## Set F · Linearisation and scaling

### F1 Linearising the pendulum

1 Square both sides. $T^2 = \frac{4\pi^2}{g} L$

2 Compare with y = m x: y is T squared, x is L, and m is 4 pi squared over g. $g = \frac{4\pi^2}{\text{gradient}}$

ANSWER B. $T^2$ against L; gradient $4\pi^2/g$

### F2 Linearising decay

1 Take natural logs of both sides. $\ln N = \ln N_0 - kt$

2 Compare with y = mx + c. $y = \ln N, \quad m = -k, \quad c = \ln N_0$

ANSWER B. ln N against t; gradient −k, intercept ln $N_0$

### F3 Scaling an orbit

1 Rearrange the proportionality for T. $T \propto a^{3/2}$

2 Scale a by 4. $4^{3/2} = (\sqrt{4})^3 = 2^3 = 8$

ANSWER 8 ×

### F4 Inverse-square scaling

1 Only r changes, so F is proportional to one over r squared. $F \propto r^{-2}$

2 Scale r by 3. $3^{-2} = \frac{1}{9} \approx 0.111$

ANSWER 0.1111 ×

### F5 Power laws

1 Take logs. $\log y = \log a + n \log x$

2 This is a straight line with y-variable log y, x-variable log x and gradient n. Social scientists will know this as the log–log (elasticity) model. $m = n, \quad c = \log a$

ANSWER C. log y against log x; gradient n

## Set G · Spot the error

### G1 Kinetic energy

1 Dividing by m happens before the root is taken, so m ends up inside the root. $v^2 = \frac{2E}{m} \Rightarrow v = \sqrt{\frac{2E}{m}}$

ANSWER B. The m should be inside the square root

### G2 Thin lens

1 You may take the reciprocal of a whole side, never of each term separately. Test with numbers: 1/2 + 1/2 = 1, but 2 + 2 is not 1. $\frac{1}{f} = \frac{u+v}{uv} \Rightarrow f = \frac{uv}{u+v}$

ANSWER B. The reciprocal of a sum is not the sum of the reciprocals

### G3 Square roots

1 Squaring destroys sign information, so the square root brings back two candidates. Context decides which to keep: a length must be positive, a displacement need not be. $x = \pm 3$

ANSWER B. The solution x = −3